# GLIDE Version 2.0: User Manual

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2026.06, Beijing

# 1. Introduction

GLIDE program is a linear inversion application for thermochronological exhumation rates modeling. The original GLIDE program was developed by Fox et al. (2014) in Fortran. Version 2.0 was subsequently rewritten in Python and optimized by Ren et al. (2026).

The basic modeling principles were described by Fox et al. (2014) and later Willett et al. (2021). As for the optimizations, see Ren et al. (2026). For the following sections, an example of the Shanxi Rift system is presented, while thermochronological and geothermal data are provided Wang et al. (2025). This manual is attached to the program.

# 2. Major functions of modeling

## 2.1 Ready for files

For a GLIDE modeling, a file of topography (.xyz) and a table of thermochronology (.csv) are required. Make sure the two files are ready for a new modeling task.

An .xyz topography file is a regular grid of elevations and contains three rows of data: longitudes (°), latitudes (°) and elevations (in meter). Positive longitudes and latitudes represent east longitude and north latitude. For the program, the points must be ranged from the northwest to east. It is worth noting that the grid of .xyz file should be larger than the modeling region. For example, the following grid (also see data\Shanxi.xyz) is established for the topography between 108.00–117.00 °E and 33.00–41.50 °N:

```
108.000000   41.500000   1616
108.010000   41.500000   1588
108.020000   41.500000   1585
108.030000   41.500000   1561
******
116.990000   41.500000   1028
117.000000   41.500000   1121
108.000000   41.490000   1491
108.010000   41.490000   1460
******
116.990000   33.000000   22
117.000000   33.000000   22
```

This file can be generated from a local DEM file (raster) or download directly through the Google Earth Engine. As for the later way, an extra function is provided in the main program and see section 3.1.

A .csv table of thermochronology data must include the six columns: “longitude”, “latitude”, “elevation”, “age”, “std” and “method”, while additional columns are allowed. Among them, “age” and “std” are the thermochronological age and its one-sigma uncertainty, using Ma as the unit. “method” contains eight major methods of thermochronology, including:

"AFT" for apatite fission track;

"ZFT" for zircon fission track;

"AHe" for apatite (U-Th)/He;

"ZHe" for zircon (U-Th)/He;

"HAr" for hornblende Ar-Ar;

"MAr" for muscovite Ar-Ar;

"BAr" for biotite Ar-Ar;

"KAr" for K-feldspar Ar-Ar.

The above eight abbreviations are not case-sensitive while handling (e.g., Aft, AhE or Mar can be recognized). Their parameters for calculating closure temperatures are given after Reiners and Brandon (2006) as listed as Table 1 ($E_a$: activation energy; Ω: a proportionality constant from diffusion parameter and effective spherical radius):

Table 1. Thermochronological parameters used in GLIDE

| method | $E_a$ (kJ/mol) | Ω ($s^{-1}$) | reference |
|---|---|---|---|
| AFT | 147 | $2.05\times10^{6}$ | Ketcham et al., 1999 |
| ZFT | 208 | $1.24\times10^{8}$ | Brandon et al., 1998 |
| AHe | 138 | $7.64\times10^{7}$ | Farley, 2000 |
| ZHe | 169 | $7.03\times10^{5}$ | Reiners et al., 2004 |
| HAr | 268 | $1.32\times10^{3}$ | Harrison, 1982 |
| MAr | 180 | 3.91 | Hames and Bowring, 1994 |
| BAr | 197 | 733 | Grove and Harrison, 1996 |
| KAr | 183 | $5.39\times10^{5}$ | Foland, 1994 |

A case of thermochronological table is as follows (also see data\Shanxi.csv; provided by Wang et al. (2025)):

```
sample, latitude, longitude, elevation, age, std, method, reference
Thm208, 38.63795494, 112.4546794, 1646, 139, 10, AFT, Cao et al. (2015)
LLS12, 37.22705, 111.22745, 1276, 84.3, 7.7, ZFT, Li and Song (2010)
FP-10, 39.0926, 114.6728, 381, 18, 1.3, AHe, Chang et al. (2019)
FP-10, 39.0926, 114.6728, 381, 70.3, 5.7, ZHe, Chang et al. (2019)
******
```

The "sample", "reference" and more properties' columns are optional but just for better management.

## 2.2 Open the program

Run GLIDE2.0.exe (or the Python code: GLIDE2.0.py) to launch the GLIDE 2.0 program (might be slow for the first time). To run the Python file, *NumPy* and *Numba* packages are required. The interface of the main program is shown in the Figure 1.

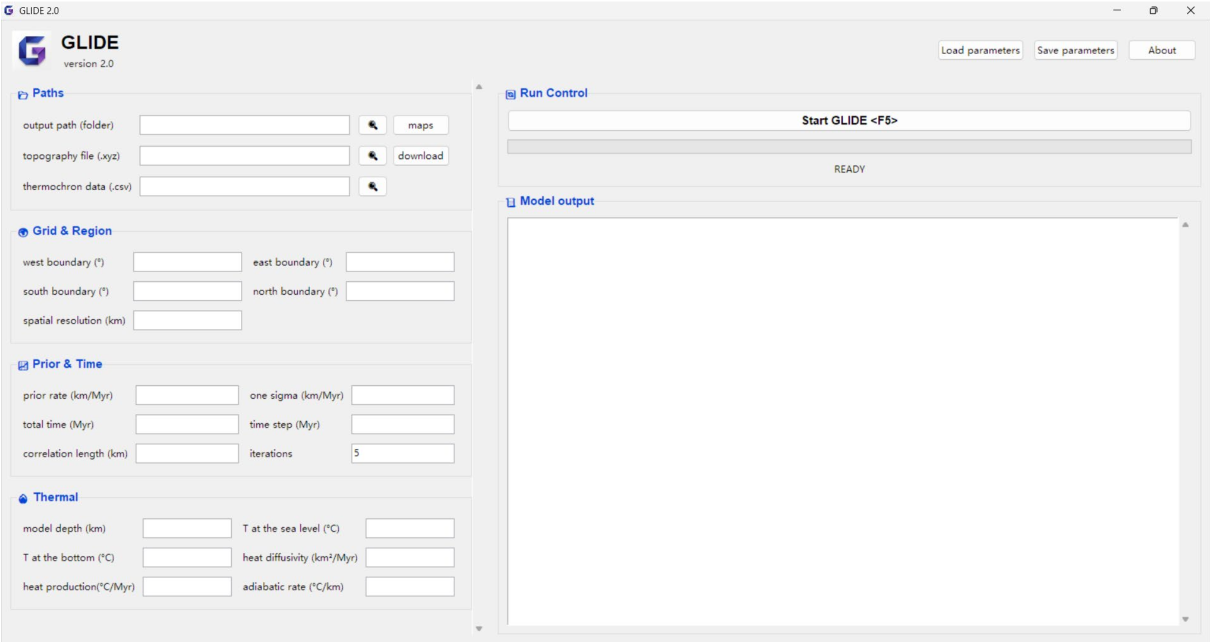


Figure 1. The interface of GLIDE main program, without inputs or outputs.

The left panel shows model related paths and parameters and the right panel shows the run control and model outputs. The upright corner present three options of load/save parameters and about (for program information).

## 2.3 Input parameters

The parameters and data used for a GLIDE modeling should be inputted in the left panel, including the following four parts.

**Paths:**

Output path: a folder; for save the outputted results by the program. See section 2.6 for details of results.

Topography file: an .xyz file; records the topographic information. See section 2.1.

Thermochron data: a .csv table; records the thermochronology dataset. See section 2.1.

**Grid & Region:**

West boundary: in degree(°); the westmost of the modeled region.

East boundary: in degree; the eastmost of the modeled region.

North boundary: in degree; the northmost of the modeled region.

South boundary: in degree; the southmost of the modeled region.

Spatial resolution: in km; the expected spatial resolution of the model results, but may be a little different from the final results.

**Prior & Time:**

Prior rate: in km/Myr (numerically equivalent to mm/yr); the prior exhumation rate of the modeled region, need to be evaluated before modeling.

One sigma: in km/Myr; the one-sigma of the prior exhumation rate, also need to be evaluated.

Total time: in Myr; the total time of modeling, i.e., time when the model start.

Time step: in Myr; the time step in modeling.

Correlation length: in km; the length used for evaluating the spatial correlations, set to 10–20 km mostly.

Iterations: the times of the inversion iterations, the default is 5 and it is stable for most cases.

**Thermal:**

Model depth: in km; the depth of the model.

T at the sea level: in °C; the temperature of the regional or adjacent sea level.

T at the bottom: in °C; the temperature of the model bottom, which decides the original geothermal gradient.

Heat diffusivity: in km²/Myr; the heat diffusivity of the model.

Heat production: in °C/Myr; the heat production expressed as an equivalent temperature increase rate, mostly set as 0.

Adiabatic rate: in °C/km; the average regional adiabatic rate, due to adiabatic effects.

After inputted, the interface will be like Figure 2., which is same to the case of Ren et al. (2026).

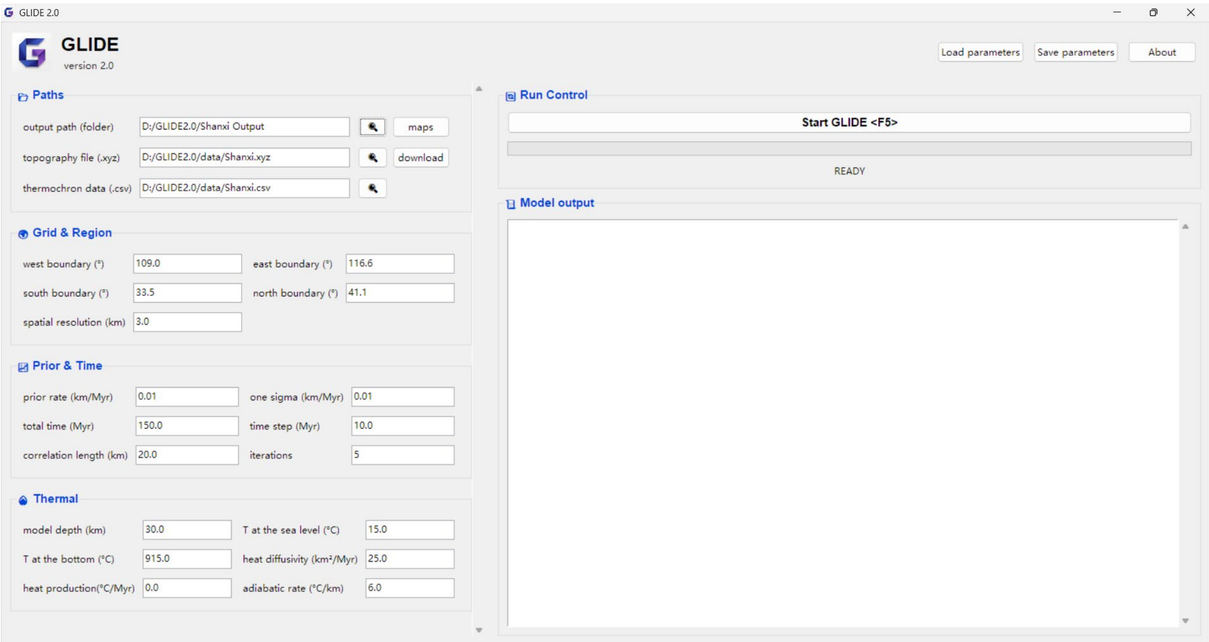


Figure 2. The interface after parameters inputted.

## 2.4 Load/save parameters

For inputting and saving the parameters more easily, the buttons “Load parameters” and “Save parameters” in the upright part can be used to read or save them into a txt file, with the shortcut key of Ctrl + O and Ctrl + S accordingly. A saved parameters’ file is (see data\Shanxi param.txt):

```
# output path (folder)
```

```
D:/GLIDE2.0/Shanxi Output
# topography file (.xyz)
D:/GLIDE2.0/data/Shanxi.xyz
# thermochron data (.csv)
D:/GLIDE2.0/data/Shanxi.csv
# west / east / south / north boundary (°)
109.0 116.6 33.5 41.1
# spatial resolution expected for model (km)
3.0
******
```

It can be edited for a new modeling, if necessary, but the lines' order cannot be changed. Meanwhile, the lines started with "#" have no effect on input.

### 2.5 Start modeling

After the preparations above, the model can be started. To run the modeling, click the "Start GLIDE" button or press shortcut key F5. And then, the progress bar will start and the state will change from "READY" to "RUNNING".

Once finished, a small window will pop up and the state will change into "FINISHED" and related output logs, including the prior misfits and $R^2$ and the posterior misfits and $R^2$ (Figure 3).

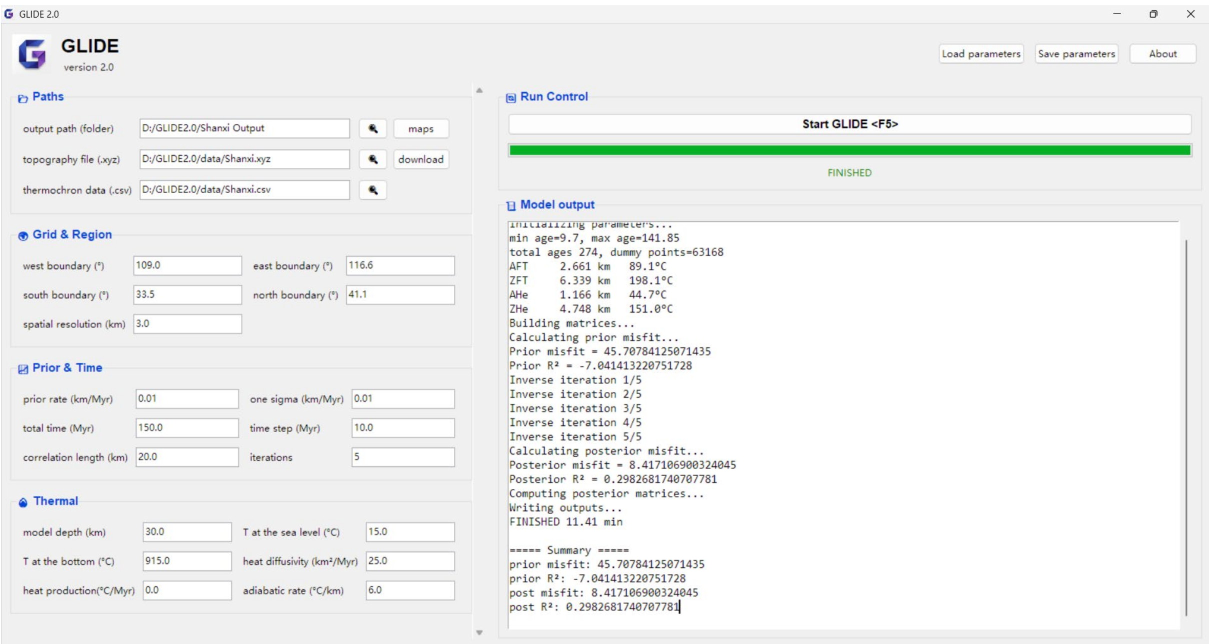


Figure 3. The interface of after modeling finished.

And the information will be shown as bellow. The misfit is defined as the weighted root-mean-square residual between observed and predicted thermochronological ages.

```
Initializing parameters...
min age=9.7, max age=141.85
total ages 274, dummy points=63168
AFT  2.661 km  89.1°C
ZFT  6.339 km  198.1°C
AHe  1.166 km  44.7°C
ZHe  4.748 km  151.0°C
Building matrices...
Calculating prior misfit...
Prior misfit = 45.70784125071435
Prior R² = -7.041413220751728
Inverse iteration 1/5
Inverse iteration 2/5
Inverse iteration 3/5
Inverse iteration 4/5
Inverse iteration 5/5
Calculating posterior misfit...
Posterior misfit = 8.417106900324045
Posterior R² = 0.2982681740707781
Computing posterior matrices...
Writing outputs...
FINISHED 11.41 min
===== Summary =====
prior misfit: 45.70784125071435
prior R²: -7.041413220751728
post misfit: 8.417106900324045
post R²: 0.2982681740707781
```

## 2.6 Results

The outputted results include a series of csv tables for distributions of posterior exhumation rate, reduced variance and time resolution, and a csv table for predicted ages. Reduced variance ranges from 0 to 1 and reflects the relative uncertainty reduction achieved by the inversion and time resolution represents the temporal resolving power of the inversion at each spatial node.

For example, the distribution of exhumation rate (in km/Myr), reduced variance and time resolution during 10–0 Ma given as (see Shanxi Output\0100.csv):

```
longitude, latitude, exhumation rate, reduced variance, time resolution
109.0,33.5,0.012700049495961188,0.9993823696200393,0.010702379104494186
109.0,33.52695034919901,0.01305541343998307,0.999210620128133,0.012106760489685899
```

```
109.0,33.553900698398024,0.013451690430919743,0.9989948860949841,0.01367173160088031
109.0,33.58085104759703,0.013892131655200389,0.9987254450891675,0.015409751155476017
******
```

And the predicted ages (in Ma) are given as (see Shanxi Output\Predicted ages.csv):

```
observed age, observed std, predicted age, method
139.0,10.0,117.78497895647415,AFT
60.0,5.0,58.36977631351052,AFT
82.0,6.0,79.34729733046484,AFT
68.0,6.0,69.56155695484094,AFT
******
```

# 3. Extra functions

There are two extra functions for downloading topography files (as .xyz) and mapping the modeled results. They are not mandatory for modeling. The both functions are optional and only provided for convenience.

## 3.1 Get topography file

An independent file Python script, download.py, is provided along with the GLIDE main program (Figure 4.). It uses the Google Earth Engine (GEE) API to allow users to download .xyz file directly, without requiring local DEM data. To run it, the numpy and earthengine-api package must be installed, and a GEE account and project must be set up. Additionally, the DEM data source can be selected from NASA, Copernicus and SRTM. If Python environment meets the requirements, simply click the "download" button in the main program to execute it; otherwise, run dowload.py directly in user's own environment.

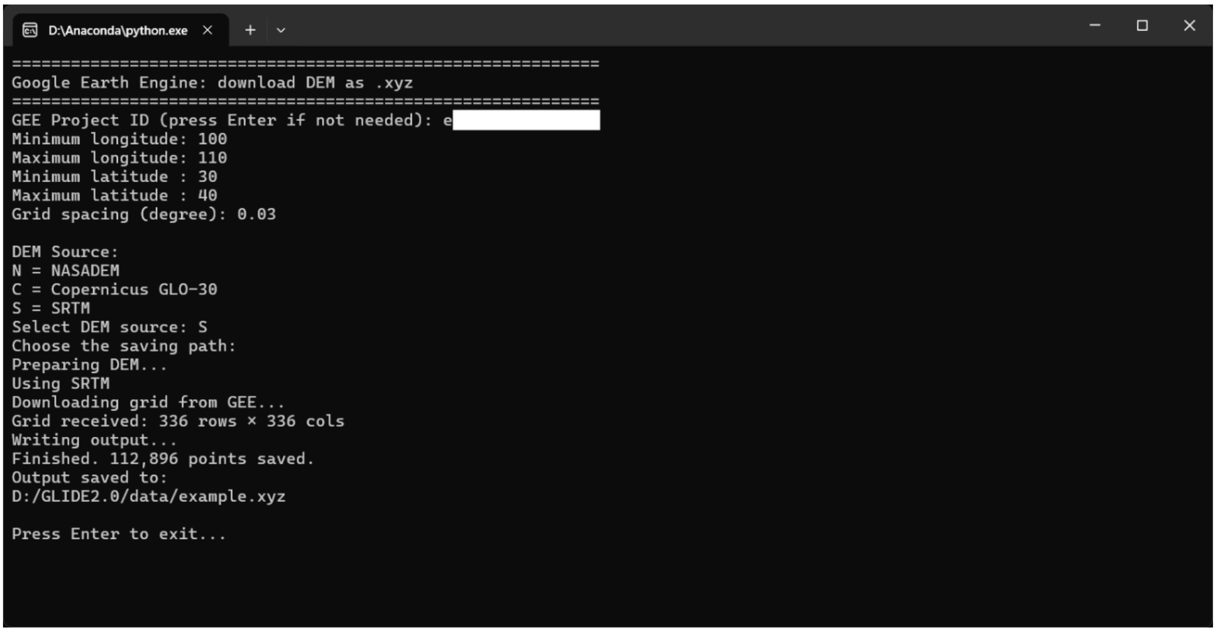


Figure 4. The interface of download.py, with inputs and outputs.

## 3.2 Map the results

An independent file Python script, mapper.py is also provided along with the GLIDE main program (Figure 5). Numpy and matplotlib package needs to be installed for running it. If Python environment meets the requirements, simply click the “maps” button in the main program to execute it; otherwise, run mapper.py directly in user’s own environment.


Figure 5. The interface of mapper.py, with inputs and outputs.

The follow interface shows the inputs for mapper code and one of maps (Figure 6, using the folder <u>Shanxi Output\</u>).

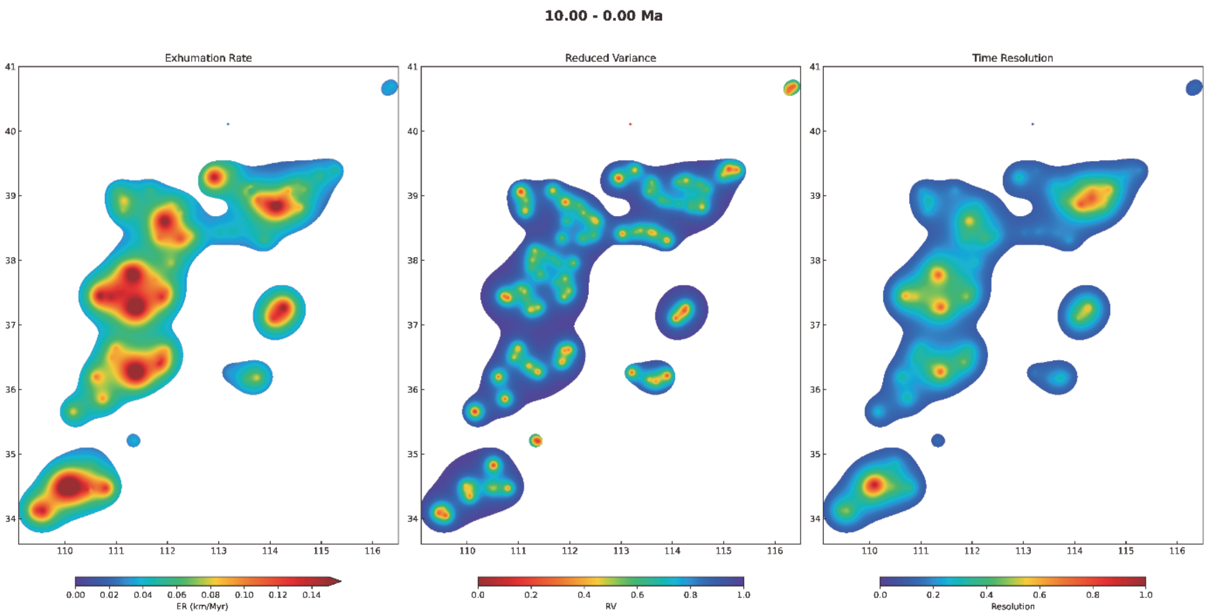


Figure 6. Generated maps for the 10–0 Ma.

## References


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